First Law Analysis of Combustion Calorimetry

ENGN 0720: Thermodynamics

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Abstract

The following report describes a bomb calorimetry experiment which was conducted using a Parr Model 1341 jacket oxygen bomb calorimeter. The goal of this experiment surrounded obtaining the heats of combustion for n-tetradecane, diesel fuel, and extra virgin olive oil. Since the calorimeter's internal components absorb some non-negligible amount of heat content, n-tetracaine, a compound with some known heat of combustion, was utilized for a calibration run. After recording the system's temperature rise, the combined heat capacity of the calorimeter system was calculated to be 10.14 kJ/K.

After the calibration run was concluded, the trials for diesel fuel and olive oil were performed, which revolved around recording the temperature change due to combustion for each sample. To obtain the most accurate results, it was important to also factor in the energy contribution of the fuse wire used for the combustion. The calculated heats of combustion of diesel and olive oil were about 44.60 kJ/g and 38.07 kJ/g respectively. These values confirmed the expectations gathered from standard HHV references for diesel fuels as well as nutritional data for olive oil. After these values were calculated, a comparison of the higher and lower heating values was conducted, validating the calorimeter's capture of the latent heat of water vapor condensation, and allowing a determination that the high heating value is the most appropriate measure of this system.

The results of this experiment successfully demonstrate that the heat of combustion for fuels and food sources can be found using the bomb calorimeter, and that small discrepancies in the final value can most likely be attributed to deviations in water mass, residual heat loss, and small differences in ignition wire length.

Introduction

Bomb calorimetry is a technique used in thermochemistry that allows close to exact measurements of heat released during a combustion reaction under a constant volume. This method is broadly used in *fuel science* as a means of quantifying hydrocarbon energy density and in *food science* to determine the caloric content of various foods and ingredients for nutrition labeling.

This experiment surrounds a constant volume combustion process, therefore, the first law of thermodynamics simplifies to $\Delta U=Q$ as there is no P-V work done on the system. A bomb calorimeter works by igniting a small sample mass in a sealed combustion chamber pressurized by oxygen. The system's temperature is raised as a result of the exothermic reaction, which serves as the basis for calculating the total heat released. The energy balance of the calorimeter system can be expressed as the following equation, where Q_{fuel} represents the heat released by

combusting the sample of fuel, Q_{wire} is the small energy addition form the burning ignition wire and C_v is the heat capacity of the bomb and water system:

$$Q_{fuel} + Q_{wire} = C_v \Delta T \quad \textbf{Eq 1}$$

During experimentation, it is important that combustion is rapid and complete. We started by charging the bomb to a high partial pressure of oxygen, ensuring a complete conversion of the fuel samples carbon and hydrogen into CO2 and H2O. The nickel alloy ignition wire (2.3 cal/cm) was used to initiate the combustion, so it became necessary to record the length of the unburned wire to determine its net heat contribution and remove that quantity from the measured total

Due to the inaccuracies which may occur as a result of accounting for every heat absorbing component of the bomb calorimeter, we conducted a calibration run using the compound n-tetradecane. Rather than individually summing all internal heat absorbing components, utilizing a compound such as n-tetradecane, where the heat of combustion is already known, allows the C_v term to easily be isolated and solved for after temperature rise is recorded during the trial run. Because of the known higher heating value of n-tetradecane, 47,351 KJ/Kg, any deviances in temperature rise from the trial run enabled us to determine the combined heat capacity of the entire system.

After C_v was determined from the calibration run, we used this value in the following trials of diesel fuel and olive oil. In each of these trial runs, we measured the mass of each fuel sample, took note of the length of fuel wire consumed, and recorded the water temperature over 15 second time intervals from just before ignition until the temperature reached a stable peak. With the final temperature rise, ΔT **,** equation one was applied and rearranged to isolate the combustion heats of the fuel sources.

Experimental Procedure

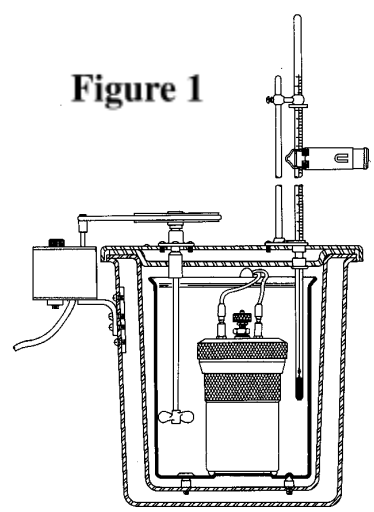


Figure 1

Apparatus:

A model 1341 jacket oxygen bomb calorimeter from Parr Instruments **(Figure 1)** was used to measure the heat of combustion of the three sample fuels (n-tetradecane [calibration], diesel fuel, and extra-virgin olive oil). The procedure was identical for each. The exterior insulating bucket is made of fiberglass and prevents any heat from escaping to the surroundings. Inside, a stainless steel bucket holds the water and the

bomb. An electrically powered propeller stirs the water and heats it up to an adequate initial temperature. A Parr 6755 Digital thermometer records the temperature of the water and outputs the readings to an external digital display. The bomb itself is a two piece conducting reaction vessel containing a sample pan for holding the fuel and attached to an ignition mechanism. It lies freely in the water bucket.

Safety:

Hazards to be kept in mind included the electrical components of the ignition system and high pressure oxygen. The following safety precautions were taken: adequate eye protection, nitrile gloves, closely following the instruction manual and careful observation of the experiment throughout.

Setup:

The first step was to fill the stainless-steel bucket with two kilograms of deionized water. A microbalance scale was used to precisely measure this quantity. After placing the water bucket into the calorimeter, the next step was to install the fuse wire to the bomb head. After clamping the head, 10cm of nickel alloy wire were cut and attached to the electrodes using tweezers. When a current is passed through, the wire ignites, along with the fuel. Subsequently, the sample pan was filled with a precisely recorded mass of fuel (between 0.75g and 1g) and installed into the bomb assembly. The end of the wire was adjusted to lie just above the fuel, without coming into contact. After carefully attaching the head and sealing, the penultimate step was to pressurize the bomb with oxygen to 25 atm using an oxygen gage and tank. Finally, the bomb was placed into the water, the electrical connections were made, and the lid was closed.

Experiment:

The propeller was switched on until a steady state water temperature was reached. Then, the ignition switch was pressed and the fuel was allowed to burn for approximately five minutes until a steady peak water temperature was reached. The digital display was recorded on a phone for the duration of the experiment. Upon completion, the calorimeter was disassembled and the complete combustion of the sample was visually verified. Any remaining length of wire was also measured. The water was emptied and the process was repeated for the remaining fuels.

Potential sources of error:

While care was taken to ensure that all measurements were made precisely, errors could have potentially arisen from lack of experience with the microbalance, spillage, or faulty equipment. Measuring the wire precisely proved to be particularly challenging. The temperature recordings were made from a reference video, and the exact timing data was likely inaccurate. Furthermore, the outer frame of the calorimeter was assumed to be a perfect insulator, but heat losses to the surroundings may have still occurred, affecting our final values. These, however, were most likely negligible.

Results

Using a heat of combustion of 45.7 kJ/g for tetradecane, a Q_{fuel} of 49.5388 kJ was calculated. Combined with a Q_{wire} of ~0.0645 kJ, this gave a Q_{total} of ~49.603 and a C_{system} of ~10.13759972.

Time	Diesel Temp (C)	Olive Oil Temp (C)	Tetra Temp (C)
0:15	22.256	22.244	22.608
5:00	26.676	26.127	27.501
Change	4.42	3.883	4.893

	Water (grams)	Fuel (grams)	Wire at start (cm)	Water produced (grams)	Wire leftover (cm)
Olive Oil	2024.12	1.032	10	0.96	3.3
Diesel	2001.76	1.003	10	1.137	2.9
Tetradecane	2009.09	1.084	10	1.45	4

Using this collected data and the calculated C_{system}, we calculate:

	Tetradecane	Olive Oil	Diesel
Wire at start (cm)	10	10	10
Wire leftover (cm)	4	2.9	3.3
$\Delta H_{combustion}$ ignition wire (cal/cm)	2.3	2.3	2.3
Conversion (kJ/cal)	0.004184	0.004184	0.004184
Q wire (kJ)	0.0577392	0.06832472	0.06447544

	Tetradecane	Olive Oil	Diesel
Q Wire (kJ)	0.0577392	0.06832472	0.06447544
Q Total (kJ)	--	39.35895396	44.80210572
Q Fuel (kJ)	49.5388	39.29062924	44.73763028

	Tetradecane	Olive Oil	Diesel
Q Fuel (kJ)	49.5388	39.29062924	44.73763028
M Fuel (g)	1.084	1.032	1.003
$\Delta H_{combustion}$ (kJ/g)	45.7	38.07231515	44.60381882

Discussion

<u>Comparison to Literature Values</u>

The expected heat of combustion (H_{comb}) for diesel falls within the range of 42-45 kJ/g, meaning our calculated value aligns well with standard results. The bottle states that 1 Tbsp of olive oil is 120 calories, giving an expected olive oil H_{comb} of ~35.86 kJ/g, which places our result of ~38.07 to be fairly similar. Our H_{comb} may be greater than expected for a number of reasons. The first is that inaccurate measurement of the fuse may have led to more of the wire being used than known, leading to additional combustion being attributed to olive oil. Secondarily, the olive oil's calorie count may have been higher than given,

leading to a lower expected H_{comb} than observed. Measurement errors with both the water and olive oil may have also played a role.

Heat Capacity Contributions

We calculated the heat in the calorimeter based only on the water and fuel, but this was inaccurate for several reasons. The calorimeter itself, including the metal container, absorbed some heat, and some heat likely escaped the system. While we factored in Q_{wire}, our measurements weren't very precise since the wire length wasn't measured exactly. That said, these factors had a minimal impact on the overall heat calculation, as the water absorbed heat on a scale three orders of magnitude higher.

In terms of the system's heat capacity, the wire and metal had negligible contributions compared to the water. While we measured the wire's small effect, the heat absorbed by the calorimeter's metal wasn't accounted for, likely leading to a slight underestimation of H_{comb}. Additionally, the water measurements weren't perfect, and the amount used varied between experiments. Depending on whether too little or too much water was used, this likely resulted in an overestimation or underestimation of H_{comb}, respectively.

Higher vs Lower Heating Value

Since the calorimeter is a closed system that captures the total energy released, including the latent heat from water vapor, HHV is the more appropriate value to use in this experiment rather than LHV. This is evident when looking at our diesel values. The calorimeter measured an HHV of approximately 44.6 kJ/g, while online sources list an HHV of 45.5 kJ/g and an LHV of 42.5 kJ/g. Our result is clearly much closer to HHV.

We can also calculate LHV values from our HHV values using a h_{vap} of 2.26 kJ/g and the equation:

$$LHV = HHV - (m_{water} \cdot h_{vap})$$

Tetradecane LHV (kJ/g)	**Olive Oil LHV (kJ/g)**	**Diesel LHV (kJ/g)**
42.423	35.90271515	42.03419882

We can see that the value for diesel here is fairly similar to the LHV value of 42.5 we found online. This further supports our conclusion that HHV is more appropriate for a calorimeter. I should note here that our mass of water was calculated by wiping the condensed water with a tissue, which is a fairly inaccurate measurement due to the accruement of other materials such as ash on the tissue, increasing the experimental mass of water. Our LHV value may have been closer to its expected value if we had measured the condensed water more accurately.

Summary

Generally, our values were within expected ranges for tetradecane, olive oil, and diesel. While our results did have some inaccuracies due to unaccounted heat absorption by the calorimeter and inconsistencies in the measurement of both the condensed and uncondensed water, their impact was minimal. We also proved experimentally that HHV was the most appropriate measure.

Conclusion

In conclusion, the bomb calorimetry lab experiment successfully determined the heat of combustion for n-tetradecane, diesel fuel, and extra virgin olive oil, validating the accuracy and reliability of bomb calorimetry for measuring the fuel and food energy content. Through calibration with n-tetradecane, we used its known heat of combustion to determine the bomb calorimeter's heat capacity, allowing for precise calculations of the heat of combustion for diesel fuel and olive oil. The results aligned similarly with standard reference values, confirming bomb calorimetry as useful for calculating a substance's heat of combustion, despite minor discrepancies likely caused by measurement errors and uncertainties in water mass and fuse wire contribution. In addition, the experiment showed that the bomb calorimeter captures total heat release, including the heat contributions from water vapor condensation (latent heat), making our results for the heat of combustion represent the Higher Heating Value of the substances tested. This experiment highlights the effectiveness of bomb calorimetry in determining the energy content of fuels and foods, demonstrating its importance in both fuel and nutrition science.

Appendix

Example error analysis calculation for olive oil:

$\Delta T = 3.883\ C$

$m_{fuel} = 1.032\ g$

$H_{combustion} = 38.072\ kJ/g$

<u>Instrumental Uncertainties</u>

Assumed uncertainty of thermometer: $\sigma_T = 0.01\ C$

Assumed uncertainty of weight scale: $\sigma_m = 0.001\ g$

<u>Temperature Percent Uncertainty:</u>

$\%\sigma_T = (\frac{0.01}{3.883}) \times 100$

$\%\sigma_T = 0.258\%$

<u>Mass Percent Uncertainty:</u>

$\%\sigma_m = (\frac{0.001}{1.032}) \times 100$

$\%\sigma_m = 0.0969\%$

<u>Add both to find heat of combustion uncertainty</u>

$\%\sigma_H = \%\sigma_T + \%\sigma_m$

$\%\sigma_H = 0.258 + 0.0969$

$\%\sigma_H = 0.3549$

<u>Absolute Uncertainty in Heat of Combustion</u>

$\sigma_H = (\frac{\%\sigma_H}{100}) \times H_{combustion}$

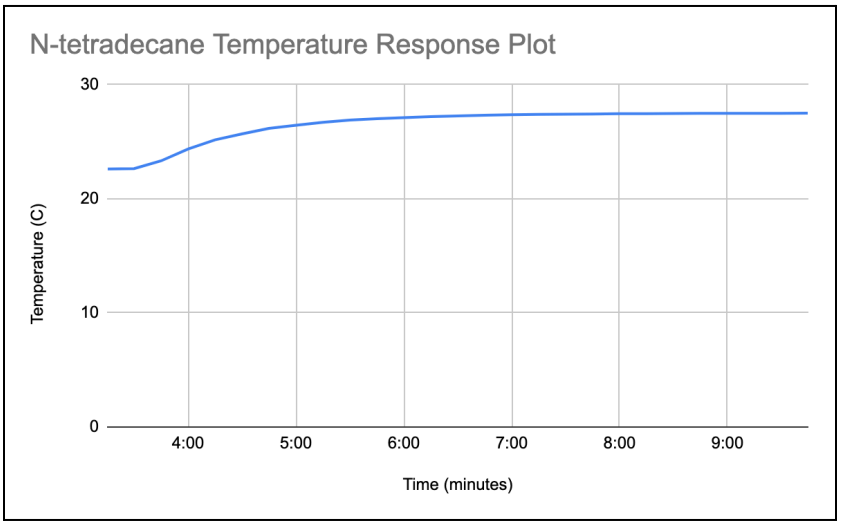
$\sigma_H = 0.003549 \times 38.072\ kJ/g$

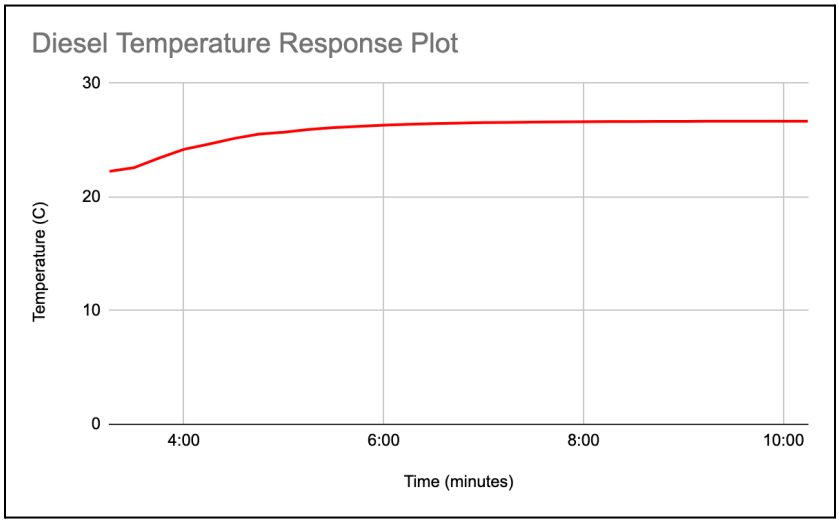
$\sigma_H = 0.13511\ kJ/g$

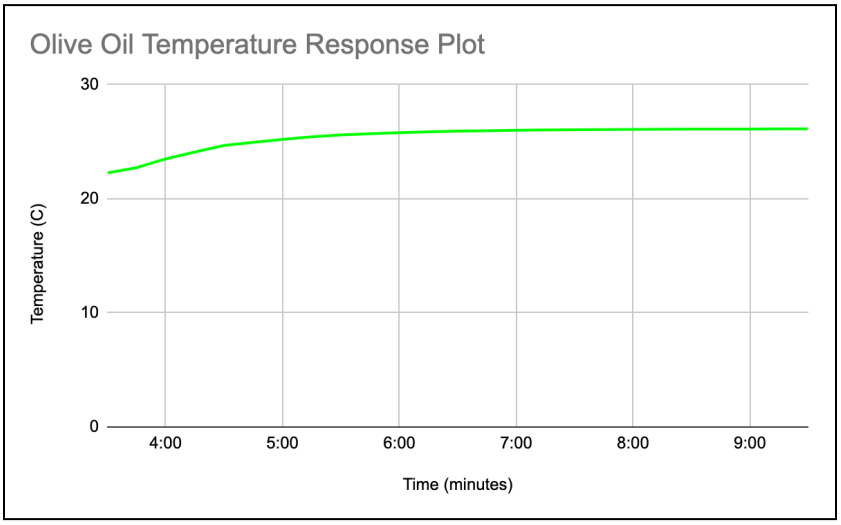
<u>Olive Oil Final Heat of Combustion with Instrumental Estimated Uncertainty</u>

$H_{combustion} = 38.072 \pm 0.13511$ kJ/g

Temperature Response Plots of n-tetradecane, diesel, and olive oil:







Example calculation of calorimeter heat capacity using n-tetradecane calibration:

Set Q of system equal to $C\Delta T$, where C is the heat capacity of the calorimeter

$Q_{system} = Q_{fuel} + Q_{wire} = C\Delta T$

N-tetradecane has a known heat of combustion, so Q fuel can be easily calculated:

$Q_{fuel} = H_{combustion} \times m_{fuel}$

$Q_{fuel} = (45.7 kJ/g)(1.084\ g)$

$Q_{fuel} = 49.5388$ kJ

Calculate heat contribution from wire

$Q_{wire} = wire\ burned\ (cm) \times H_{combustion}\ (cal/cm)$

$Q_{wire} = (6\ cm)\ \times\ (2.3\ cal/cm)$

$Q_{wire} = 13.8\ cal$

$Q_{wire} = 13.8\ cal\ \times\ \frac{0.004184\ kJ}{1\ cal} = 0.0577\ kJ$

Find Q total and solve for C

$Q_{total} = Q_{fuel} + Q_{wire} = C\Delta T$

$49.5388\ +\ 0.0577\ =\ C\ *\ (4.893\ Celsius)$

$C\ =\ 10.14\ kJ/K$